



LOST AND FOUND CAMPUS TRACKER

A smart, location-aware recovery network for the campus community

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Campus lost & found is slow, scattered, and untrustworthy

1

No central record

Reports live across notice boards, WhatsApp groups, and word of mouth — nothing is searchable in one place.

2

Manual, error-prone matching

Students must personally scan every board or group to spot a match, and easily miss one.

3

No location insight

There's no way to see where items are commonly lost, so recurring hotspots go unnoticed.

4

Weak ownership proof

Claims are settled by conversation alone, opening the door to disputes and false claims.

02 · EXISTING SYSTEM

Today's process is manual and disconnected

- Physical notice boards and lost-and-found counters at scattered campus locations
- Informal WhatsApp / social media groups used to share lost or found items
- Purely manual searching — students and staff compare reports by memory or by browsing posts
- No matching logic — potential matches are found only if someone happens to notice
- No location tracking, no analytics on where items are commonly lost
- No ownership verification mechanism — claims rely on trust alone
- No notification system — users must keep checking back manually

Proposed System

A centralized, intelligent recovery platform for the whole campus

1

Centralized reporting

Students & staff log lost/found items with category, description, location, date/time and photo.

2

Smart matching engine

A rule-based algorithm compares reports and returns potential matches with a confidence score.

3

Location-aware tracking

A campus map highlights frequently reported areas via the Google Maps API.

4

Private verification

Ownership is confirmed using identifying details hidden from general users.

5

Automatic notifications

Email alerts fire the moment a found item closely matches an open lost report.

6

Recovery lifecycle

Every report moves through
Reported → Possible Match →
Verification Pending → Verified →
Recovered.

04 · OBJECTIVES

What the system sets out to achieve

1 Build one centralized recovery trail for every lost or found item on campus

2 Automate matching with a rule-based algorithm and a confidence score

3 Provide location-aware tracking and a campus hotspot map

4 Protect ownership through private, verified identification

5 Deliver real-time notifications the moment a possible match appears

6 Give administrators tools to moderate, resolve disputes and manage duplicates

05 • SCOPE

Scope of the project

IN SCOPE

- + Web application for students & staff to report lost / found items
- + Structured reports: category, description, location, date/time, image
- + Rule-based matching engine with confidence scoring
- + Campus map with location-based hotspot tracking
- + Private ownership verification workflow
- + Email notifications on possible matches
- + Admin panel for moderation & duplicate management
- + Recovery lifecycle status tracking

OUT OF SCOPE (Phase 1)

- Native mobile applications (Android / iOS)
- Physical courier or item-transport service
- Payment, reward or bounty system
- Integration with campus ID/security systems
- AI-based deep-learning image recognition (only optional image-assisted classification is considered)

06 · TECHNOLOGY IDENTIFICATION

Technology stack

FRONTEND

HTML5, CSS3, JavaScript, Bootstrap

BACKEND

Java with Spring Boot

DATABASE

MySQL

DATA ACCESS

Spring Data JPA / Hibernate

MATCHING ENGINE

Rule-based algorithm comparing category, description, location & date

LOCATION SERVICES

Google Maps API

NOTIFICATIONS

Email notification service for match alerts

DEV TOOLS

VS Code, IntelliJ IDEA, Postman

VERSION CONTROL

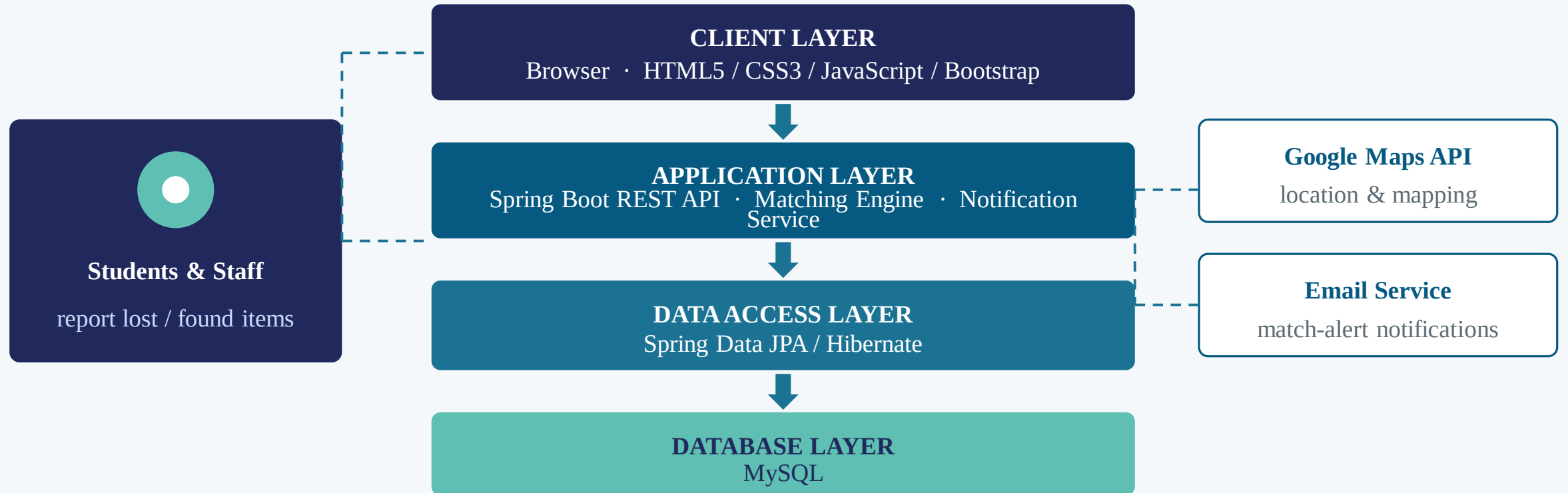
Git & GitHub

DEPLOYMENT

Render / Railway

07 · BASIC ARCHITECTURE

System architecture



Thank You